

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I D.Raghu Vardhan 192311051_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

d.raghu vardhan

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Learning Platform

System Components:

Course Module, Video Streaming, Quiz Module, User Authentication

Given Constraints:

- Video loading time < 3 seconds
- Secure login with OTP
- System must support 1000+ concurrent users

Question:

Analyze the system using **quality assurance techniques** under the given constraints. Identify possible defects and suggest testing strategies to ensure performance and security. Justify your approach. (BL4)

Q1. Online Learning Platform – Quality Assurance Analysis (BL4)

Introduction

The Online Learning Platform consists of four major components: **Course Module, Video Streaming Module, Quiz Module, and User Authentication Module**. The system must satisfy the following constraints:

Video loading time < 3 seconds

Secure login with OTP authentication

Support for 1000+ concurrent users

Quality Assurance (QA) ensures that the system meets both functional and non-functional requirements through systematic testing and defect prevention techniques.

1. Quality Assurance Analysis

A. Performance Requirement: Video Loading Time < 3 Seconds

Possible Defects

Slow video buffering.
High server response time.
Network latency issues.
Large video file size.
Poor caching mechanisms.

Testing Strategies

Performance Testing

Measure video loading speed under normal and peak conditions.

Load Testing

Check system behavior with many users accessing videos simultaneously.

Stress Testing

Evaluate system performance beyond expected user limits.

Caching Verification

Test whether frequently accessed videos are loaded from cache.

Expected Result

Video starts loading within 3 seconds.
No buffering delays under normal load.

B. Security Requirement: OTP-Based Login

Possible Defects

OTP delivery failure.
OTP expiration errors.
Unauthorized login attempts.
Weak password storage.
Session hijacking vulnerabilities.

Testing Strategies

Authentication Testing

Verify username, password, and OTP validation.

Security Testing

Check protection against:
SQL Injection
Cross-Site Scripting (XSS)
Session Hijacking
Brute Force Attacks

OTP Validation Testing

Verify:
OTP generation
OTP expiration

Single-use OTP functionality

Penetration Testing

Simulate cyberattacks to identify security weaknesses.

Expected Result

Only authenticated users gain access.

OTP expires after specified time.

Unauthorized access is prevented.

C. Scalability Requirement: 1000+ Concurrent Users

Possible Defects

Server crashes.

Slow response times.

Database bottlenecks.

Session management failures.

Resource exhaustion.

Testing Strategies

Load Testing

Simulate 1000+ concurrent users.

Scalability Testing

Increase users gradually and observe system behavior.

Endurance Testing

Run the platform continuously for long periods.

Database Performance Testing

Evaluate query execution and transaction handling.

Expected Result

System remains stable.

No significant performance degradation.

User requests are processed successfully.

2. Component-wise Quality Assurance

Course Module

Possible Defects

Incorrect course information.

Broken course links.

Missing content.

Testing

Functional Testing

UI Testing

Integration Testing

Video Streaming Module

Possible Defects

Video playback failure.

Audio-video synchronization issues.

Buffering delays.

Testing

Performance Testing

Compatibility Testing

Load Testing

Quiz Module

Possible Defects

Incorrect score calculation.

Question loading errors.

Submission failures.

Testing

Functional Testing

Boundary Value Testing

Database Testing

User Authentication Module

Possible Defects

Invalid OTP acceptance.

Login failures.

Session timeout issues.

Testing

Security Testing

Authentication Testing

Penetration Testing

3. Quality Assurance Techniques Used

QA Technique	Purpose
Functional Testing	Verify system functions correctly
Performance Testing	Measure speed and responsiveness
Load Testing	Evaluate behavior under expected user load
Stress Testing	Identify breaking point of system
Scalability Testing	Ensure support for future growth
Security Testing	Detect vulnerabilities
Penetration Testing	Simulate cyber attacks
Integration Testing	Verify module interaction
Regression Testing	Ensure changes do not create defects

4. Justification of the Approach

The selected QA techniques directly address the system constraints:

Performance Testing ensures videos load within the required 3 seconds.

Security and Penetration Testing verify OTP-based authentication and prevent unauthorized access.

Load and Scalability Testing confirm the platform can support more than 1000 concurrent users.

Integration and Regression Testing ensure all modules work together correctly after updates. This combination provides complete coverage of functionality, performance, security, and scalability requirements.

Conclusion

To ensure the success of the Online Learning Platform, Quality Assurance must focus on **performance, security, and scalability testing**. Performance testing validates fast video delivery, security testing protects OTP-based authentication, and load/scalability testing ensures support for 1000+ concurrent users. By applying these QA techniques, the platform becomes reliable, secure, efficient, and capable of delivering a high-quality learning experience.

ONLINE LEARNING PLATFORM



